Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/05/2006
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	The Melting of Aqueous Foams		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	08/05/2004	**End Date:**	06/30/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GB61G		
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Task Description:	The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology. There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into ASL technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability. The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ASL, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam.
Task Progress:	During the last year we made significant progress in three areas: [1] Graduate student Mr. Alex Gittings and I investigated bubble motion in a coarsening foam using speckle-visibility spectroscopy. In this method, which we introduced and are still developing, we characterize motion by the “visibility” of the speckle pattern formed by laser light transmitted or backscattered from a sample. If the bubble motion is slow during the exposure of a camera, then the speckle is less visible. This effect can be quantified and interpreted in terms of dynamics. The great advantage over usual techniques is that it opens up the possibility of measuring motion that changes with time. For example, we use it to measure the duration and rate of sudden stick-slip avalanche-like rearrangements of bubbles from one packing configuration to another. We obtain the full distributions of these quantities. By contrast, diffusing-wave spectroscopy only gives the average rate of rearrangements. In our case, the rearrangement dynamics are driven by the aging process. Similar dynamics occur during flow, and change systematically as a function of strain rate. [2] Dr. Pamela Korda and I worked toward an apparatus for measuring the full three-dimensional structure of dry foams. Our approach is to perform tomography using unscattered light. This is more challenging than conventional computed x-ray tomography, where photons are either absorbed or transmitted ballistically, because in our case photons are either scattered or transmitted ballistically and hence the scattered light can reach the detector and interfere with the signal. Thus, modern methods still have not been applied to measuring bubble topology statistics for bulk foams consisting of very large numbers of bubbles. MRI and photography have been applied to small samples, but the record for best statistics still belongs to Matzke who visually inspected monodisperse foams in the 1940’s. We hope to beat this record and collect good statistics for coarsening polydisperse foams, which are more generic. We hope to create a device that is sufficiently rapid as to allow measurement of dynamics, as well. To date we are able to reconstruct small samples and have identified the technical hurdle that must be overcome in order to reconstruct larger samples. Dr. Korda has written up her progress in an informal report, and has now taken a job with a biotech startup firm. [3] Dr. Klebert Feitosa and I worked on producing and characterizing steady-state foams, in which gas bubbles are created at constant rate underneath a pool of surfactant solution. This give rise to a column of foam bubbles, with small wet bubbles at the bottom and large dry bubbles at top. We measure the liquid fraction vs height via electrical conductivity. We measure bubble size and vertical speed via optical imaging. We analyze the results in terms of both the drainage equation and the coarsening equation. These specify, respectively, the liquid / bubble speed and the rate of change of bubble size as a function of bubble size and liquid fraction. In the past, these have been used individually for drainage issues, where coarsening is absent, and for coarsening issues, where drainage is absent. Now we test them more severely, and look for new effects, by examining a well-controlled situation where both processes are present. This is an important step toward understanding and ultimately controlling multiphase separations, where of course all these processes play a role. In addition to this work, Dr. Feitosa and I are examining the fundamental underpinnings of electrical conductivity as a probe of liquid fraction.
Bibliography Type:	Description: (Last Updated: 06/12/2025)
Articles in Peer-reviewed Journals	Gittings AS, Bandyopadhyay R, Durian DJ. "Photon channelling in foams." Europhysics Letters. 2004 Feb;65(3): 414-9. http://dx.doi.org/10.1209/epl/i2003-10084-4 , Feb-2004

Articles in Peer-reviewed Journals	Bandyopadhyay R, Gittings AS, Suh SS, Dixon PK, Durian DJ. "Speckle-visibility spectroscopy: A tool to study time-varying dynamics." Review of Scientific Instruments 2005 Sep;76(9):093110. http://dx.doi.org/10.1063/1.2037987 , Sep-2005
Awards	Durian DJ. "Fellow of the American Physical Society, March 2005." Mar-2005
Awards	Durian DJ. "Election as Member at Large for APS Topical Group on Statistical and Nonlinear Physics, December 2004." Dec-2004
Presentation	Gittings, A.S.; Durian, D.J. "Speckle Visibility Spectroscopy and Bubble Rearrangements" 2005 March Meeting of the American Physical Society Mar-2005
Presentation	Feitosa, K.; Durian, D.J. "Coarsening driven transport of gas in steady-state foam" 2005 March Meeting of the American Physical Society Mar-2005
Presentation	Korda, P.; Durian, D.J. "Three-Dimensional Visualization of an Aging Foam" 2005 March Meeting of the American Physical Society Mar-2005
Presentation	Durian, D.J. "The Optics of Foams" 5th European Conference on Foams, Emulsions and Applications Jul-2004